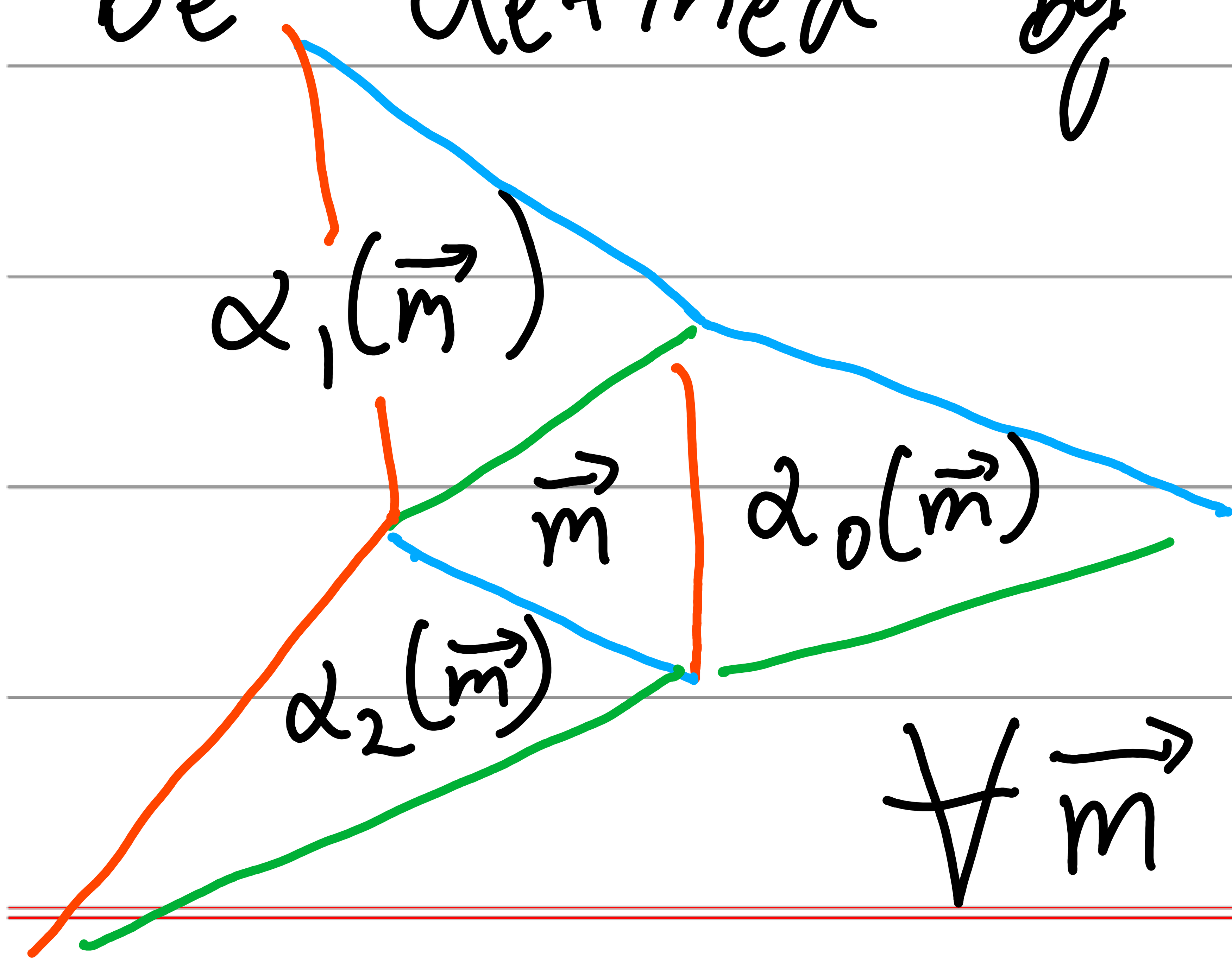
Constructing Surfaces

Choose a triple of orientation-reversing Möbius involutions $\vec{\iota} = (\iota_0, \iota_1, \iota_2)$.

For $i \in \{0,1,2\}$, let $\alpha_i : \hat{\mathbb{R}}^3 \to \hat{\mathbb{R}}^3$

be defined by $\alpha_i(\vec{m})_j = \begin{cases} \iota_i(m_j) & \text{if } i \neq j \\ m_j & \text{if } i = j. \end{cases}$



$G = \langle \alpha_0, \alpha_1, \alpha_2 \rangle$,

$\mathcal{O} \subset \hat{\mathbb{R}}^3$ a G-orbit.

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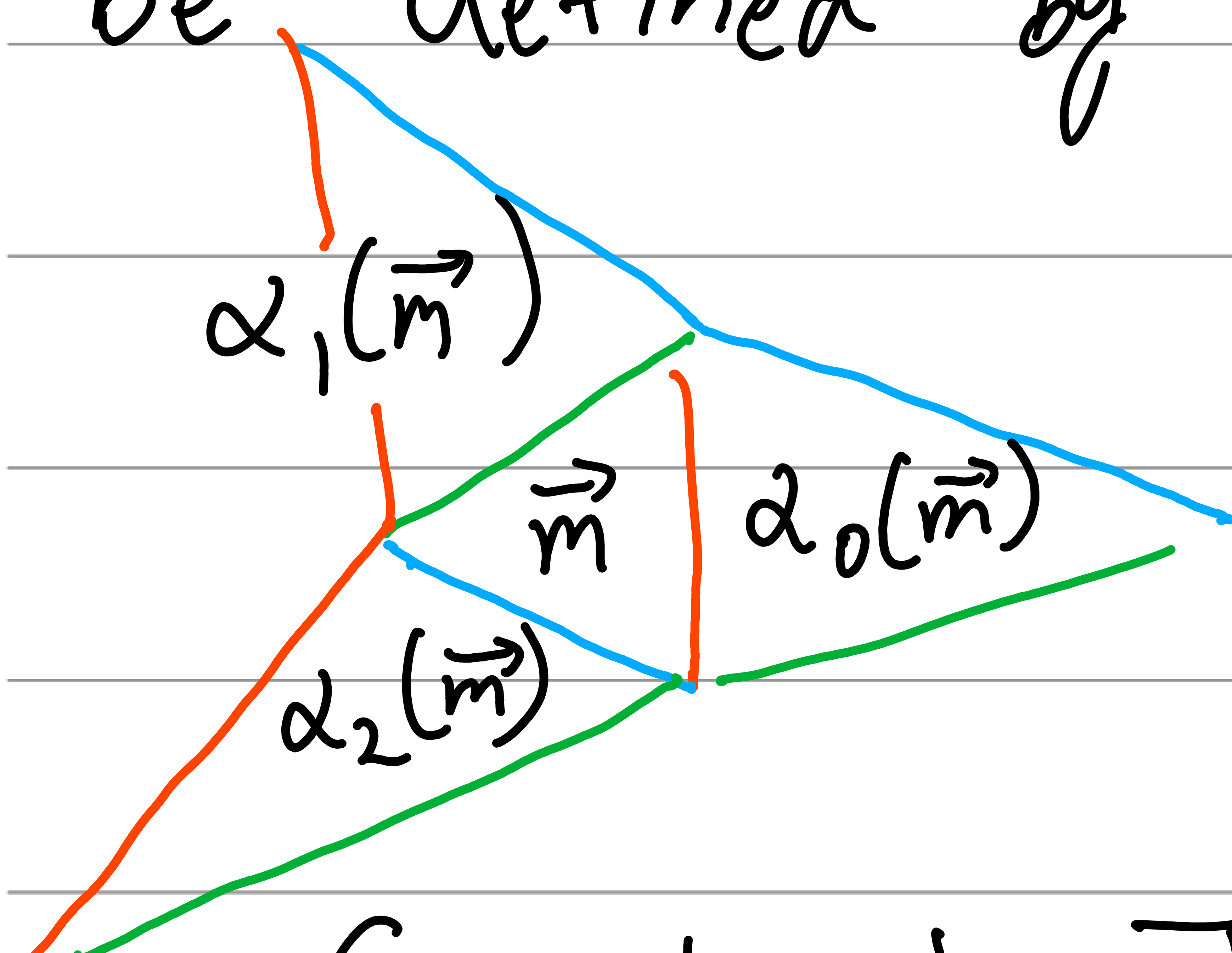

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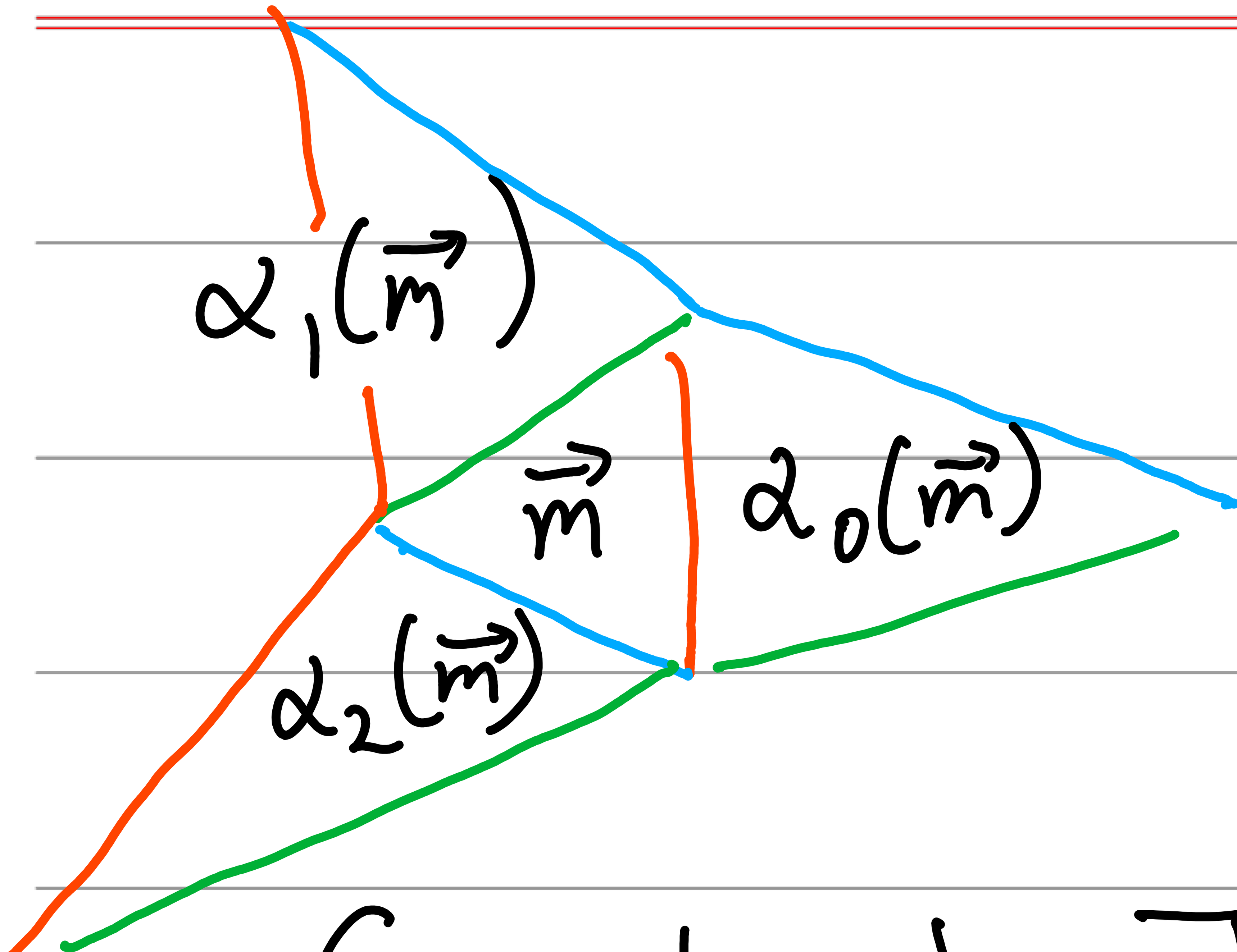
Glue edge i of $T_{\vec{m}}$ to edge i of $T_{\alpha_i(\vec{m})}$.



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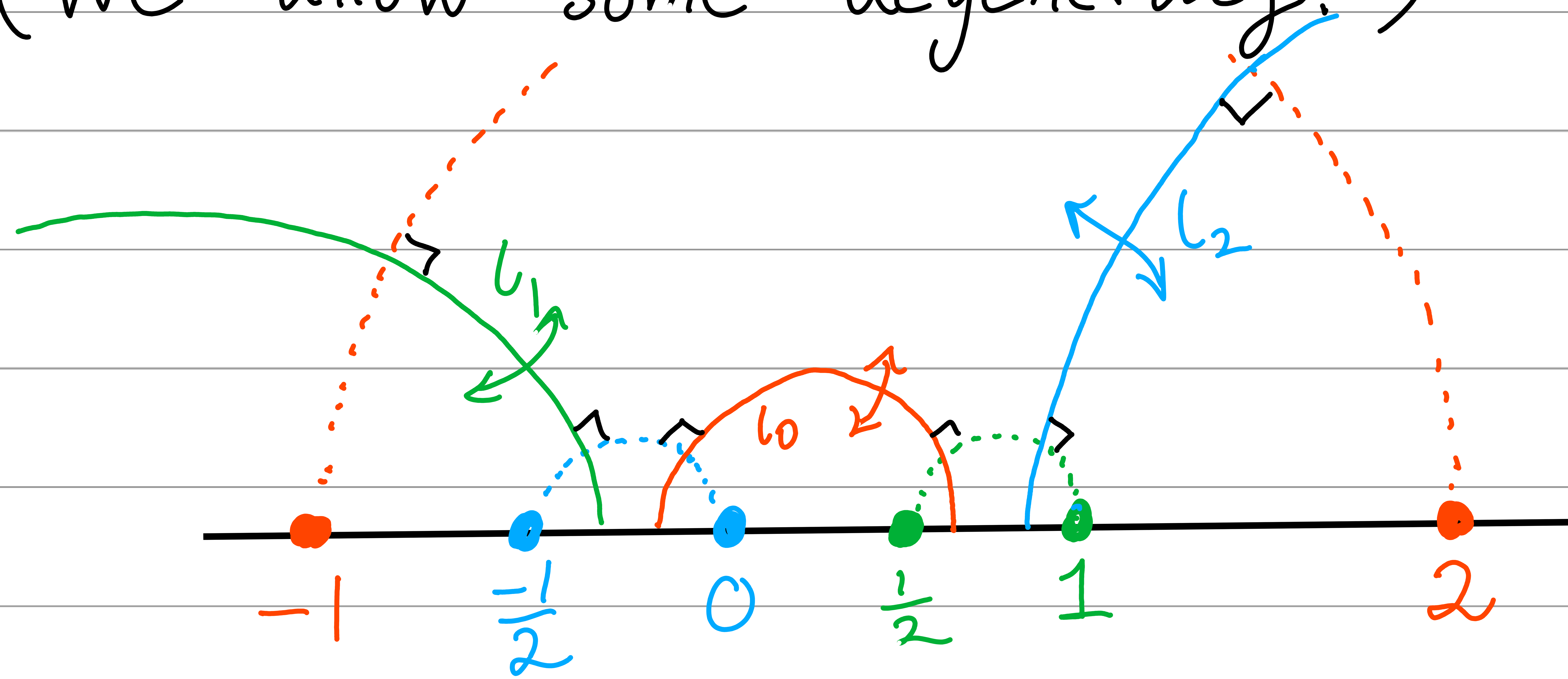
Ex

Red: 0
Green: 1
Blue: 2



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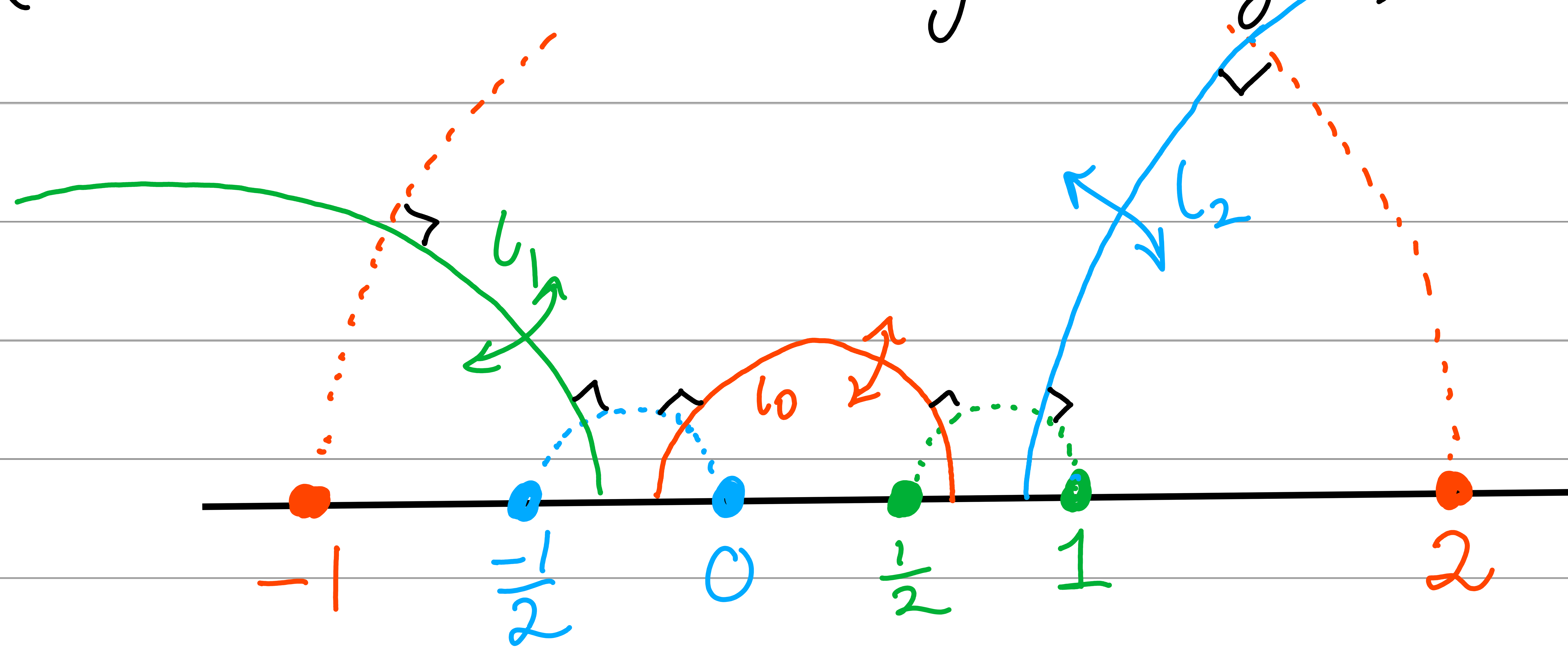

This construction will be successful because all triples (red, green, blue) decrease.

Ex

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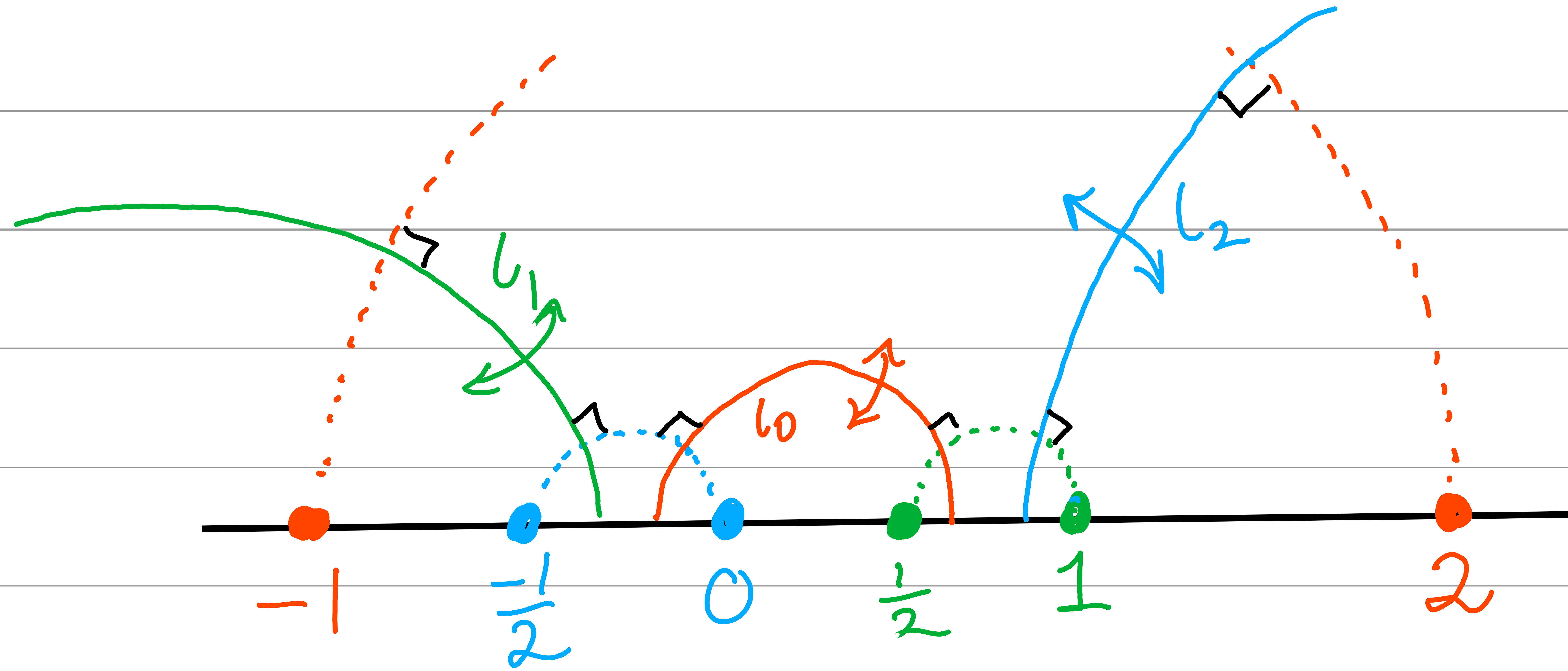
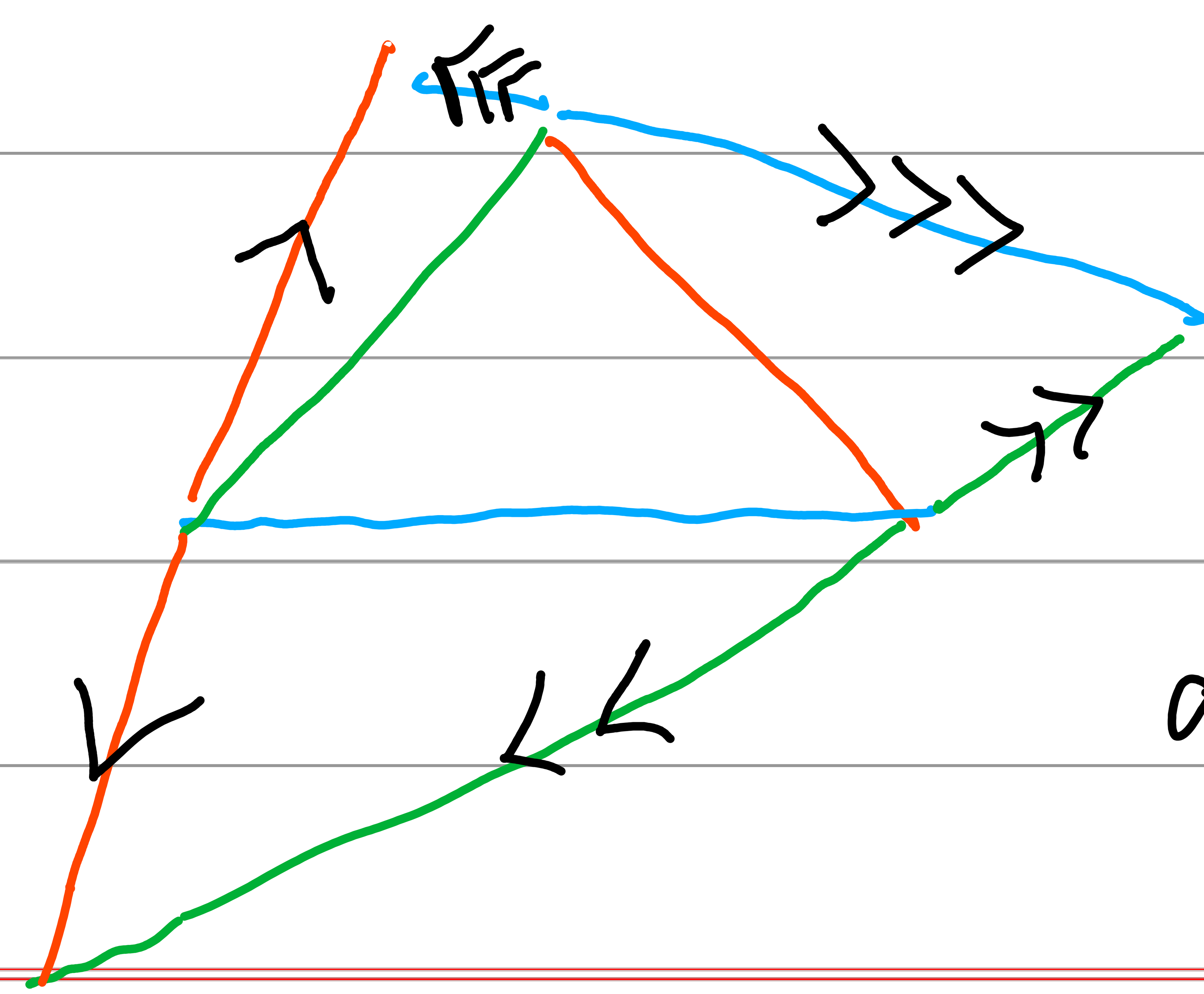


$\mathcal{O} = G(-1, 0, 1)$. $S = S(\vec{\iota}, \mathcal{O})$:



Studied by Taro Shima

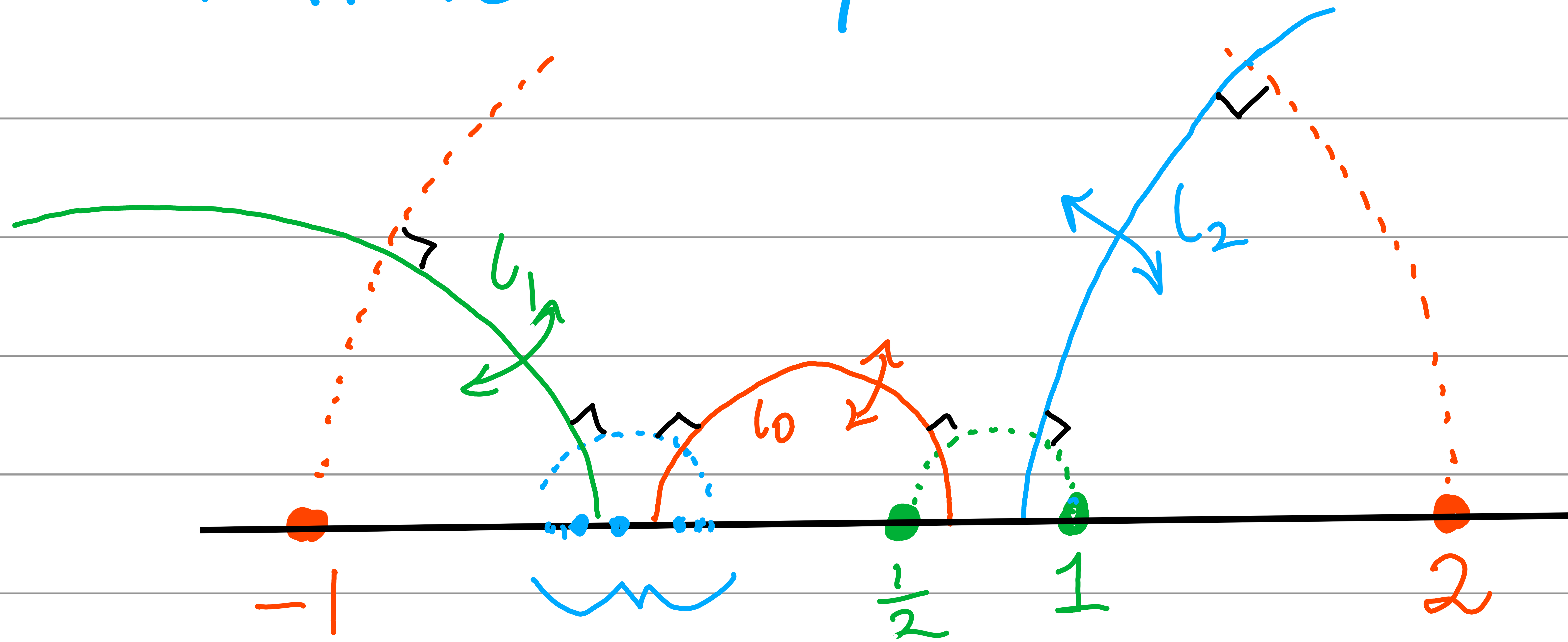
arXiv:2205.00100

Related infinite example:

Red: 0

Green: 1

Blue: 2



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Thm If θ is finite, the orbit $PSL(2,\mathbb{R}) \cdot S$ is closed in the space of triangulable half-dilation surfaces.

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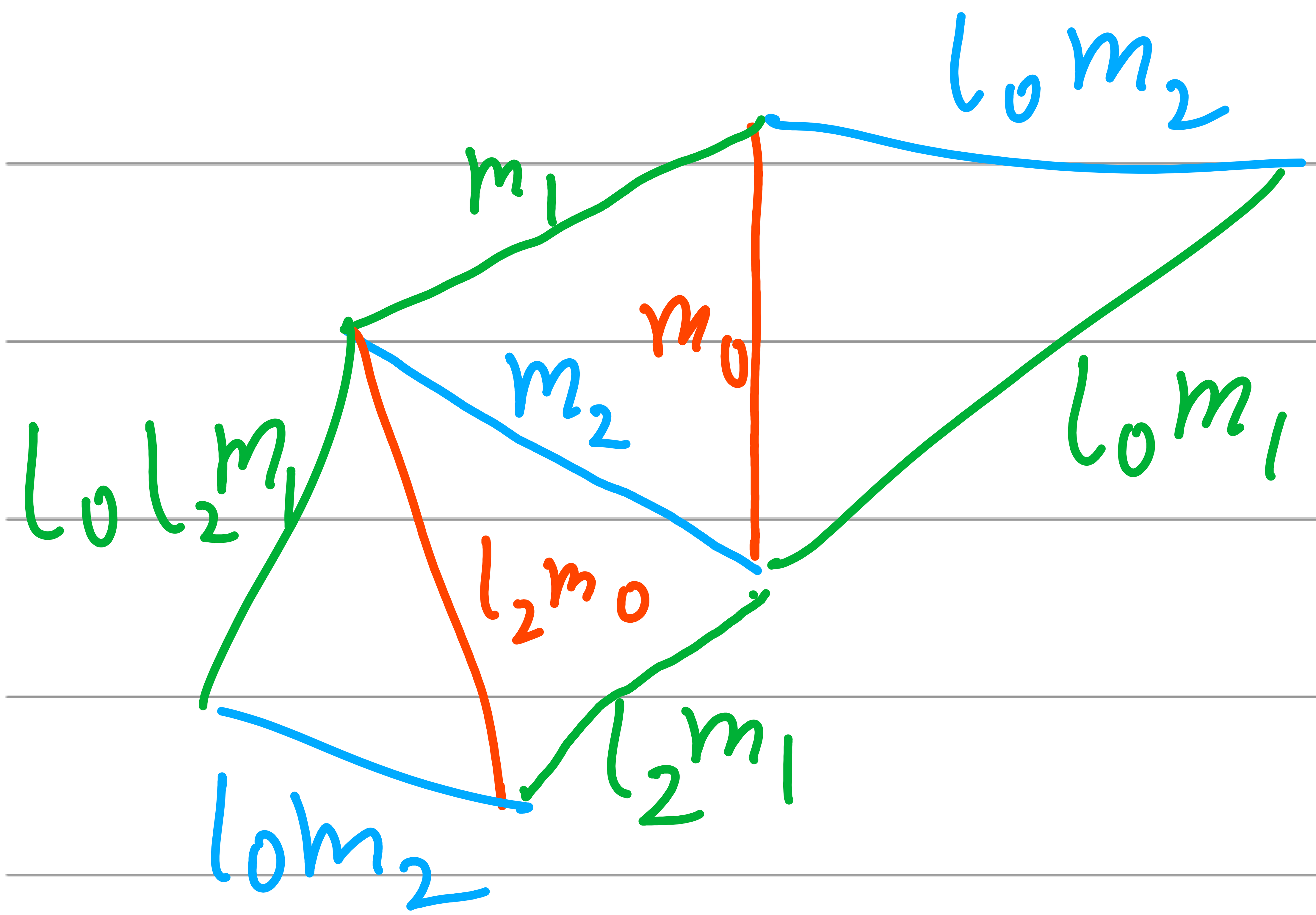
Lemma If we flip all edges with index 0, we go from $S((\iota_0, \iota_1, \iota_2), \theta)$ to $S((\iota_0, \iota_0\iota_1\iota_0, \iota_0\iota_2\iota_0), (\iota_0, I, \iota_0) \cdot \theta)$.

Lemma If we flip all edges with index 0, we go from $S((l_0, l_1, l_2), \sigma)$ to $S((l_0, l_0 l_1 l_0, l_0 l_2 l_0), (l_0, l_0, I)\cdot \sigma)$.



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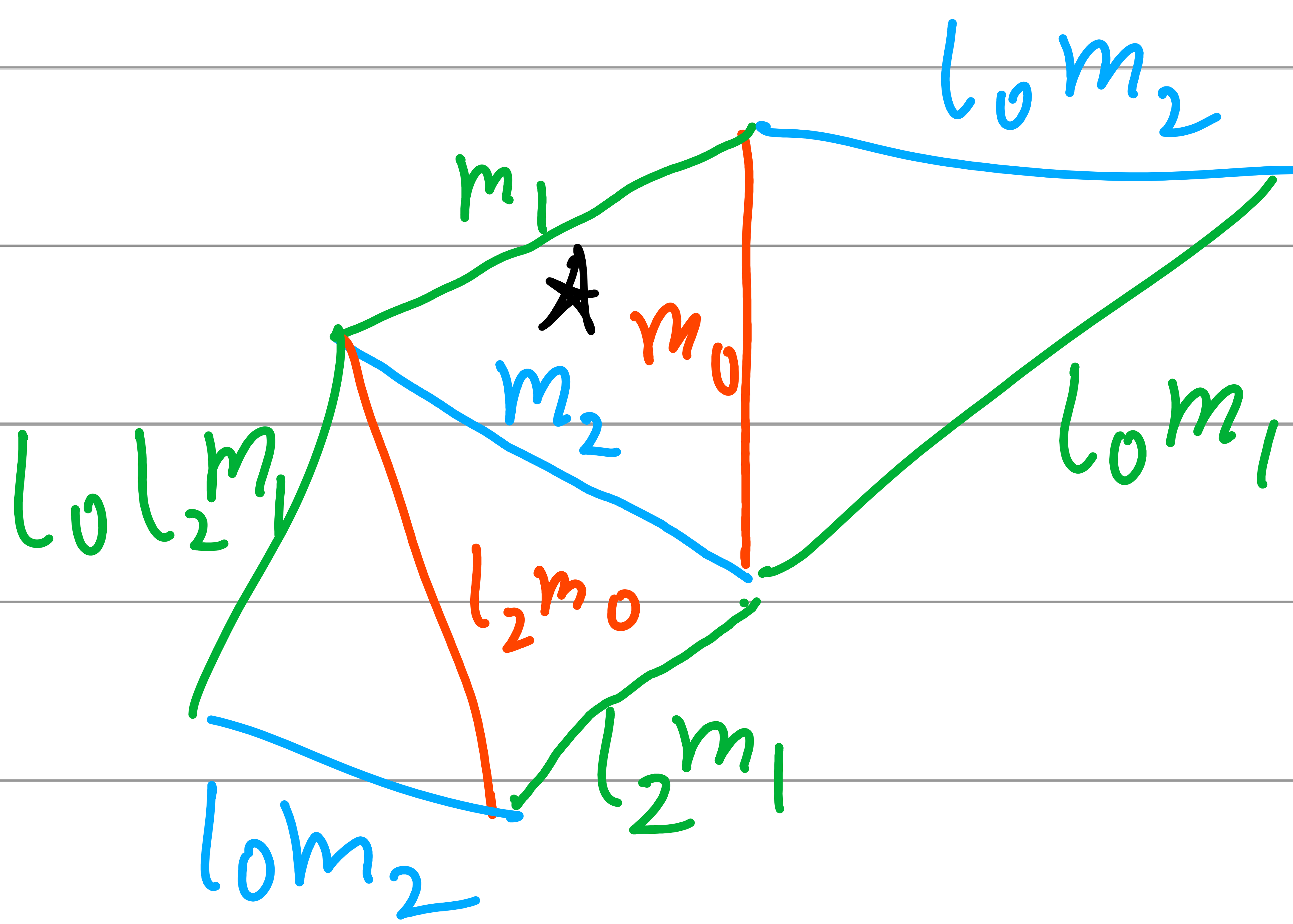
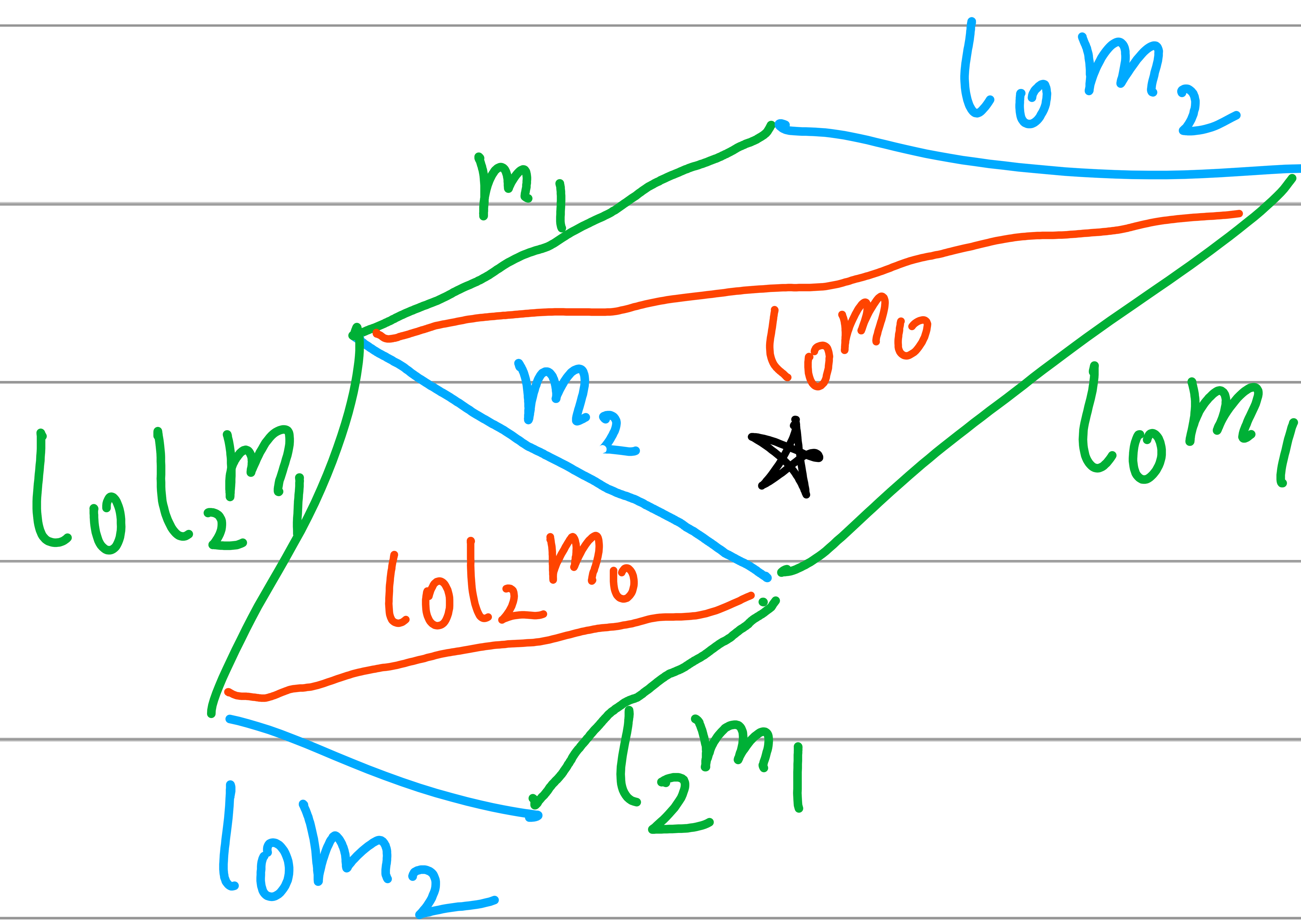
$$(l_0, l_0, I)\cdot\sigma = (l_0, I, l_0)\cdot\sigma$$





Lemma If we flip all edges with index $\mathcal{O}$, we go from $S = S((\iota_0, \iota_1, \iota_2), \mathcal{O})$ to $S_0^{\circ} = S((\iota_0, \iota_0\iota_1\iota_0, \iota_0\iota_2\iota_0), (\iota_0, \iota_0, I)\cdot\mathcal{O})$.

Thus, $\iota_0 \cdot S = \iota_0 \cdot S_0^{\circ} = S(\vec{\iota}, (I, I, \iota_0)\cdot\mathcal{O})$

Lemma If we flip all edges with index 0, we go from $S = S((\iota_0, \iota_1, \iota_2), \theta)$ to $S_0' = S((\iota_0, \iota_0\iota_1\iota_0, \iota_0\iota_2\iota_0), (\iota_0, \iota_0, I)\cdot\theta)$.

Thus,

$$\iota_0 \cdot S = \iota_0 \cdot S_0' = S(\vec{\iota}, (I, I, \iota_0)\cdot\theta)$$

$$\iota_1 \cdot S = \iota_1 \cdot S_1' = S(\vec{\iota}, (\iota_1, I, I)\cdot\theta)$$

$$\iota_2 \cdot S = \iota_2 \cdot S_2' = S(\vec{\iota}, (I, \iota_2, I)\cdot\theta)$$

Lemma If we flip all edges with index $\mathcal{O}$, we go from $S = S\big((\iota_0, \iota_1, \iota_2), \mathcal{O}\big)$ to $S_0' = S\big((\iota_0, \iota_0\iota_1\iota_0, \iota_0\iota_2\iota_0), (\iota_0, \iota_0, I)\cdot\mathcal{O}\big)$.

Thus,

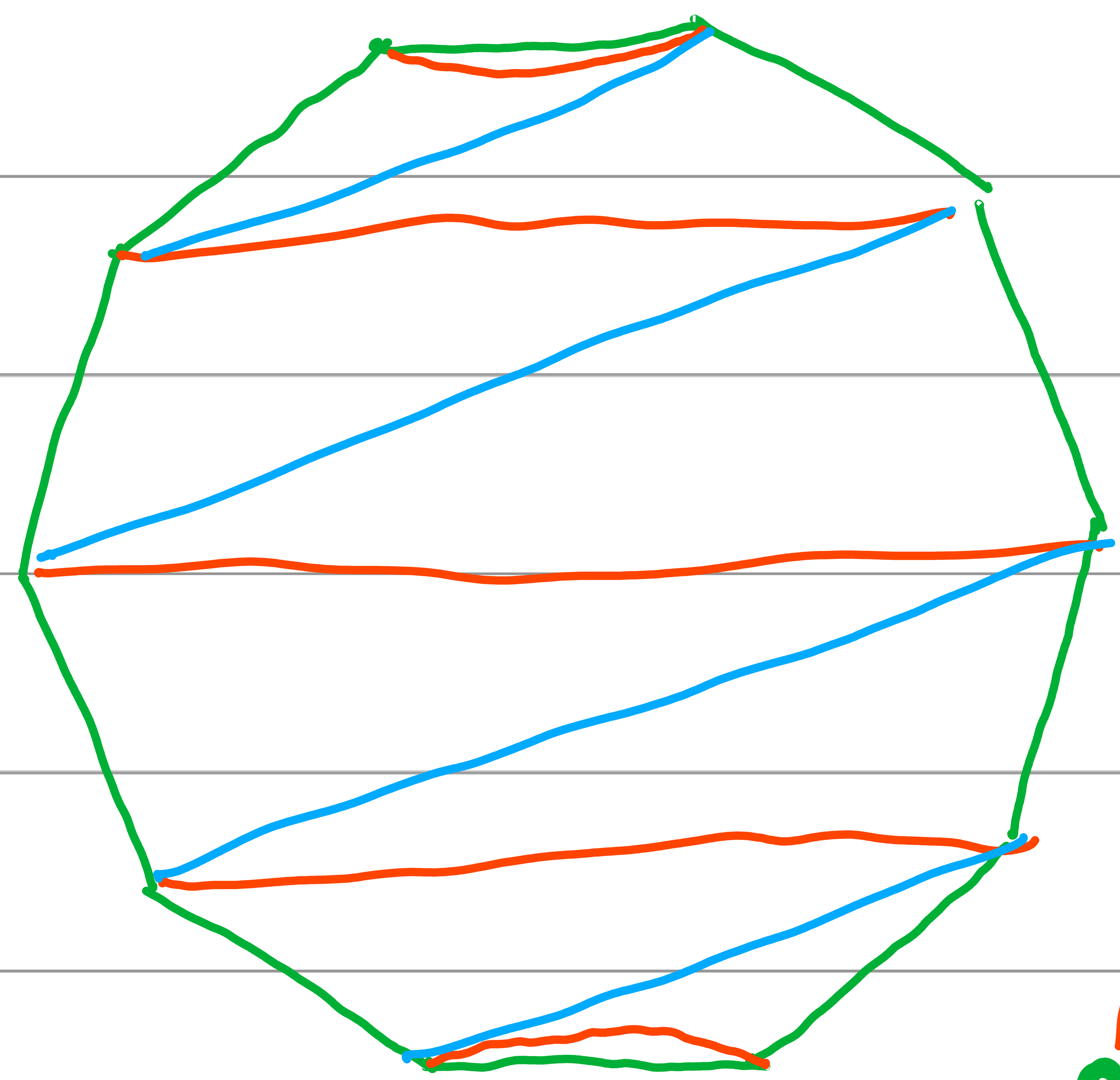
$$\iota_0 \cdot S = \iota_0 \cdot S_0' = S\big(\vec{\iota}, (I, I, \iota_0)\cdot\mathcal{O}\big)$$

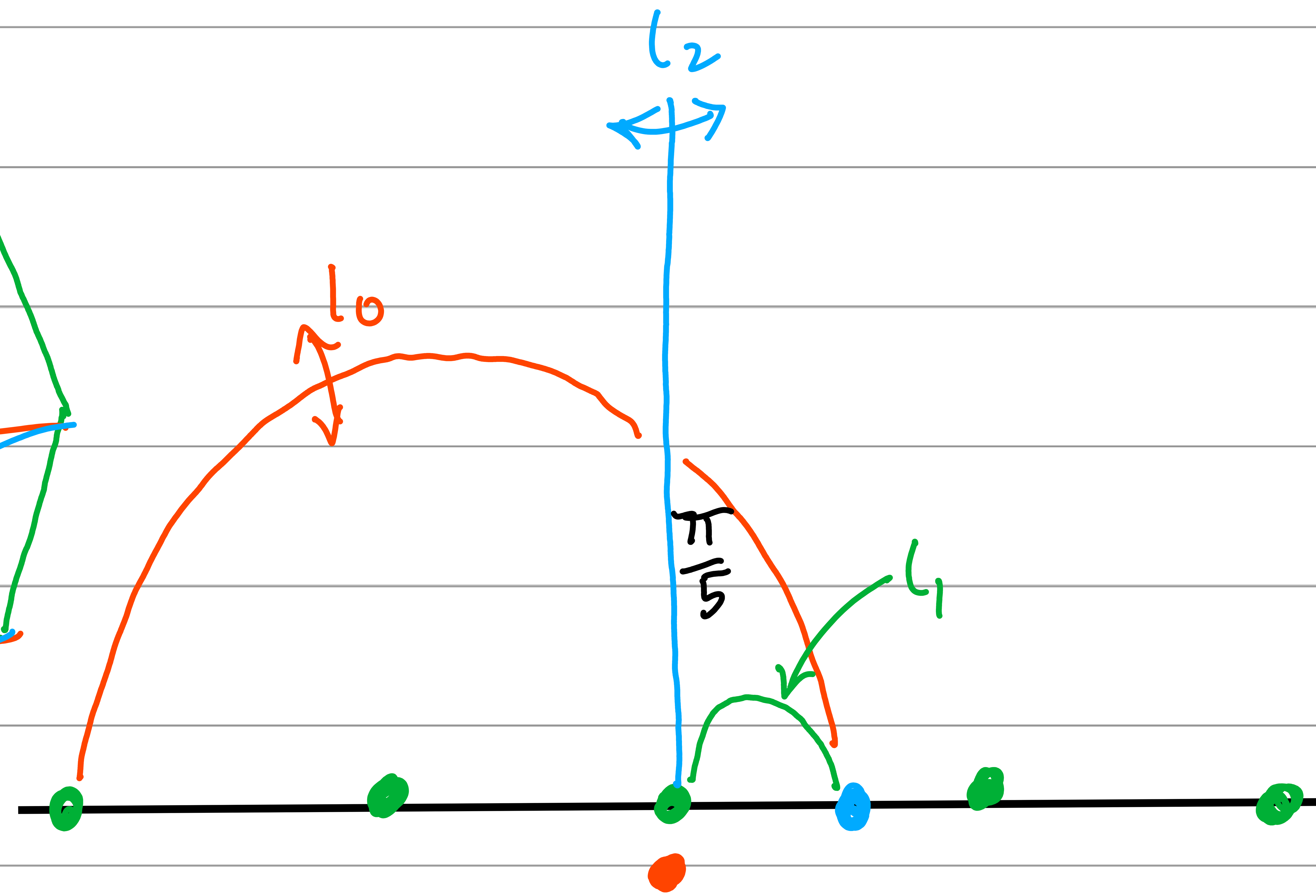
$$\iota_1 \cdot S = \iota_1 \cdot S_1' = S\big(\vec{\iota}, (\iota_1, I, I)\cdot\mathcal{O}\big)$$

$$\iota_2 \cdot S = \iota_2 \cdot S_2' = S\big(\vec{\iota}, \underbrace{(I, \iota_2, I)}_{(\mathbb{Z}/2\mathbb{Z})^3\text{-action}}\cdot\mathcal{O}\big)$$

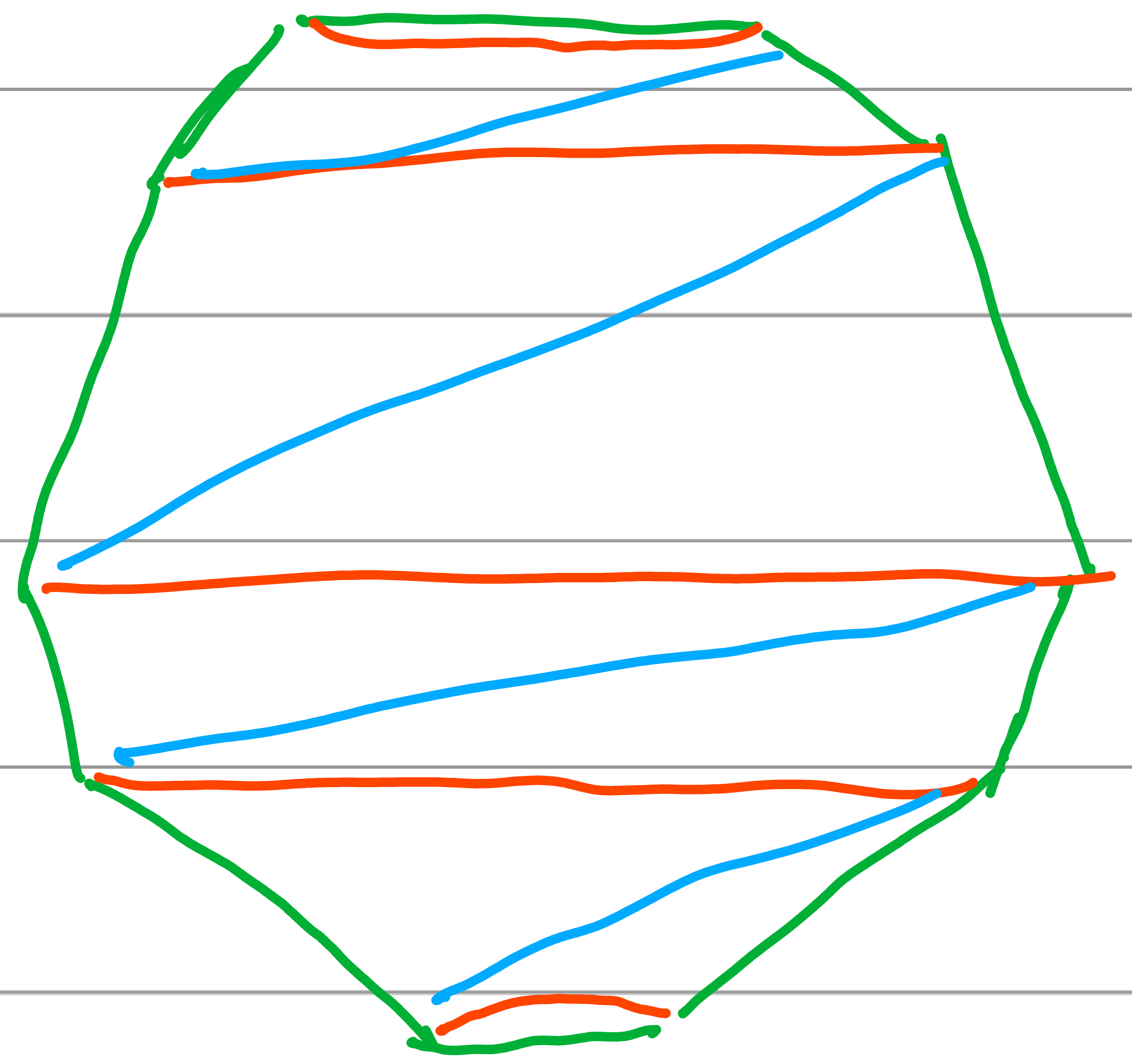
Ex Deforming Veech's Surfaces.

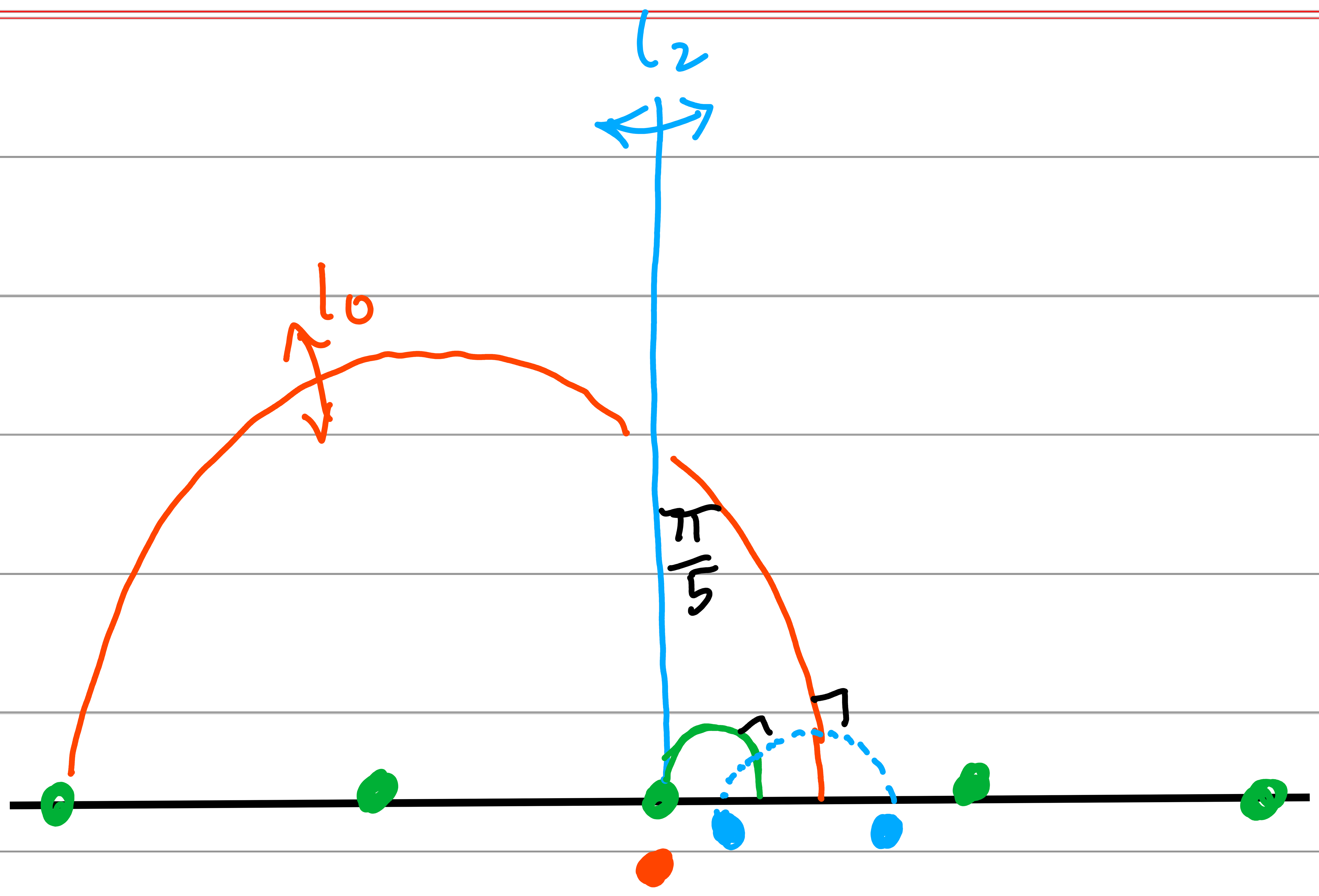
Regular 10-gon



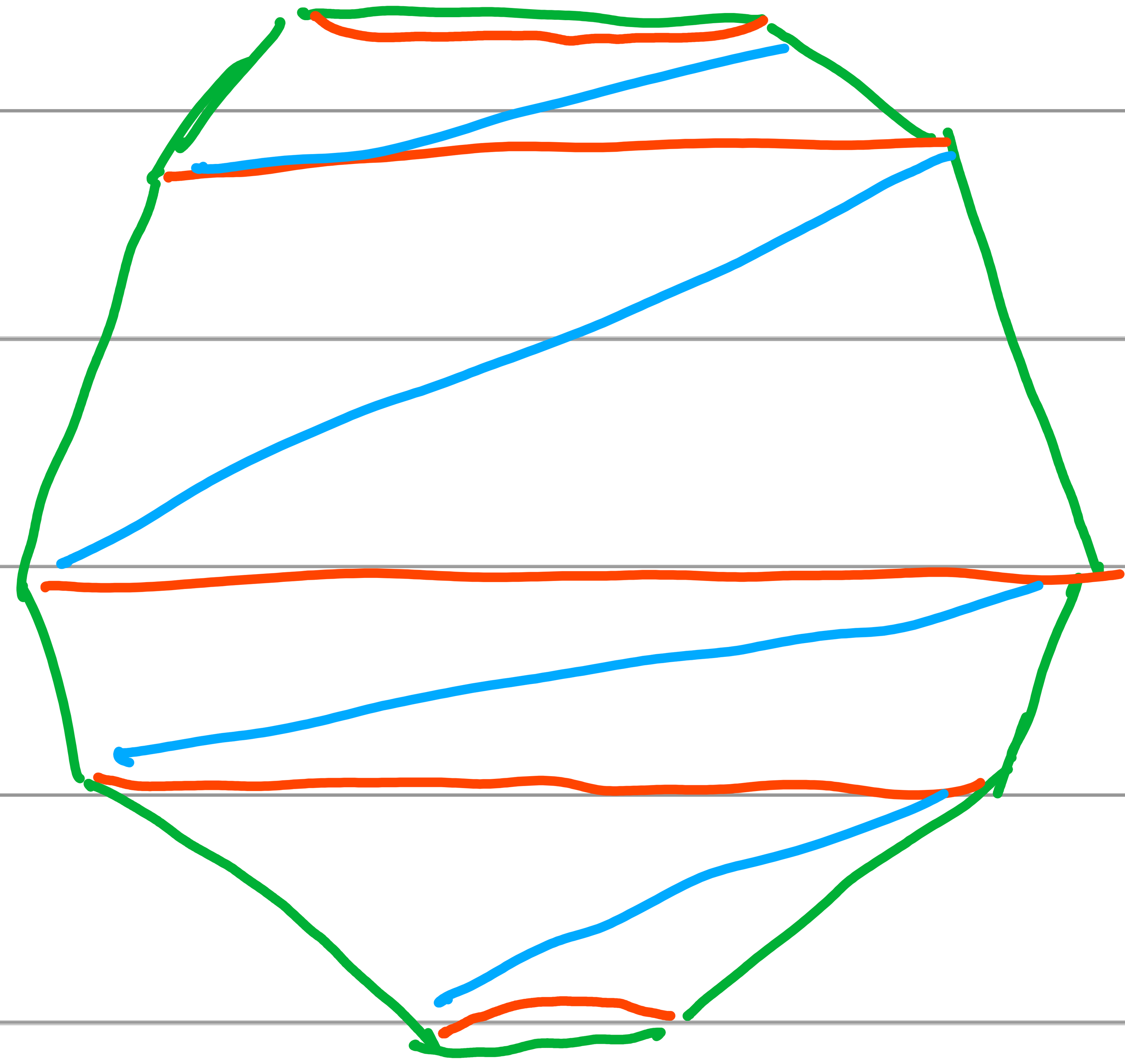


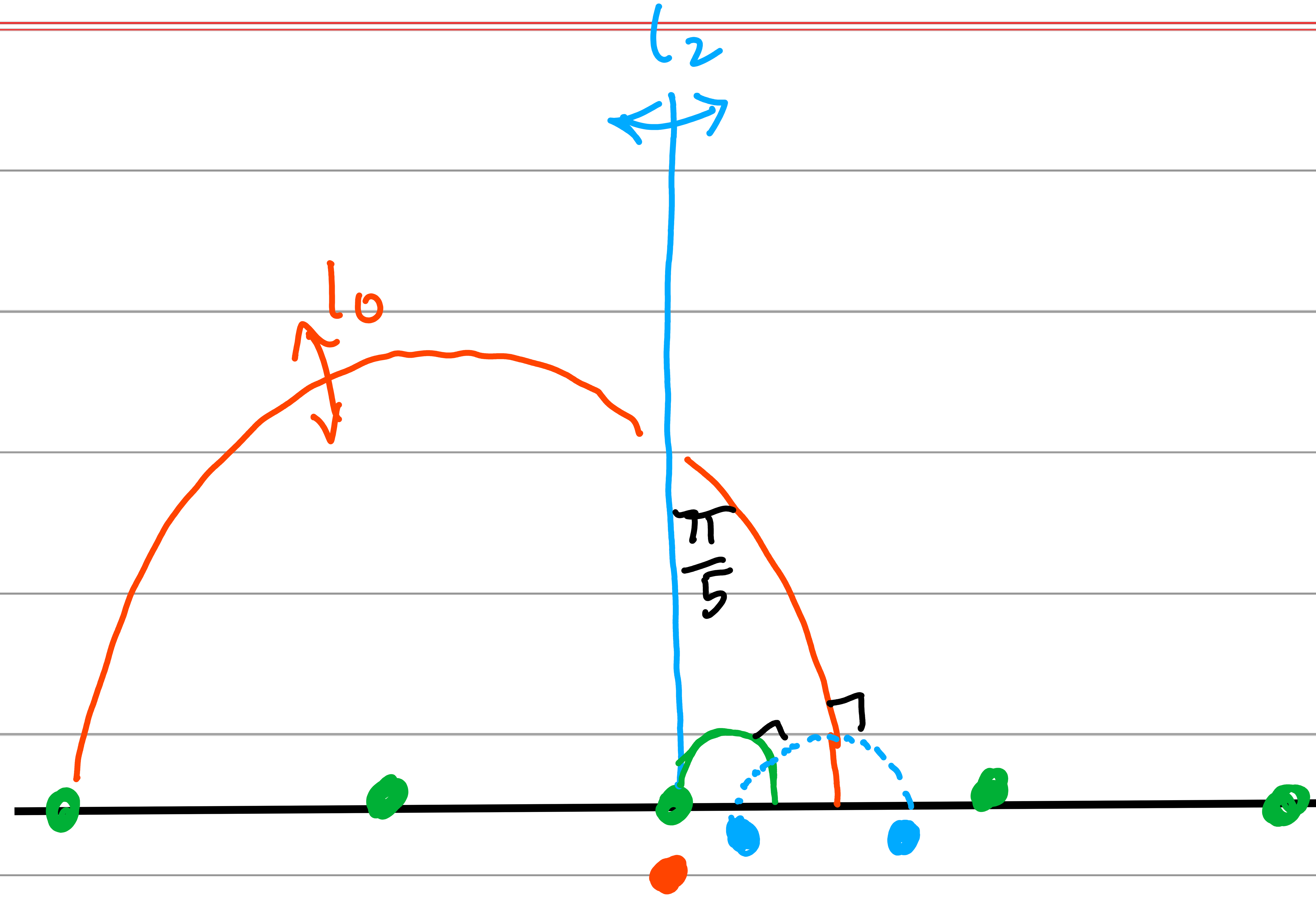
degenerate





l_2
l_0
$\frac{\pi}{5}$





Infinite variant